INCH-POUND

MIL-C-82283C(NU)

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SUPERSEDING
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MILITARY SPECIFICATION

CLOTH, TWILL, COTTON, LIGHTWEIGHT

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a lightweight cotton twill cloth.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, P.O. Box 59, Natick, MA 01760-0001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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SPECIFICATIONS

FEDERAL

V-T-276 - Thread, Cotton
PPP-P-1134 - Packaging, Packing of Cotton and Cotton-Synthetic Fiber-Blend Fabrics (Excuding Duck Fabrics)

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspections by Attibutes
* MIL-STD-2073-1 - DOD Materials Procedures for Development and Application of Packaging Requirements
* MIL-STD-2073-2 - Packaging Requirement Codes

* (Unless otherwise indicated, copies of federal and military specification, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

* 2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of documents which are DoD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

*TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

Method No. 8-1988 - Colorfastness to Crocking: AATCC Crockmeter Method

(Applications for copies should be addressed to AATCC National Headquarters, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material.

3.3.1 Yarn. The yarn shall be made of cotton which has been cleaned, carded, drawn, and spun into singles for both the warp and the filling when tested as specified in 4.4.1 and 4.5.

3.4 Physical requirements. The finished cloth shall conform to the requirements in Table I when tested as specified in 4.5.

Table I - Physical Requirements

Weight oz per sq yd (g/m^2)		Yarns per inch (minimum)(cm)		Breaking Strength lbs (min)(Newton)	
Min.	Max.	Warp	Filling	Warp	Filling
5.5 (186.5)	6.0 (203.4)	72 (28.3)	58 (22.8)	50 (222.5)	60 (267)

3.4.1 Weave. The weave shall be a 2/1 right or left hand twill when tested as specified in 4.5.

3.4.2 Width. The width of the cloth shall be as specified (see 6.2), and shall be the minimum width, inclusive of selvage when fly shuttle looms or shuttleless with tuck-in selvage looms are used. For all other shuttleless looms, the width measurement shall be made between the last warp yarn on each side excluding the protruding fringe(s).

3.5 Color. Unless otherwise specified (see 6.2), the finished cloth shall be dyed to match Olive Green 107.

3.5.1 Sulfur content. The use of sulfur dyes or dyes containing elementary sulfur or sulfur compounds capable of oxidation to sulfuric acid is prohibited. The finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.5. When a standard sample is not available, the dyed cloth shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.5.

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* 3.5.2 Matching. The color and appearance of the dyed and finished cloth shall match the standard sample when viewed under filtered tungsten lamps which approximate artificial daylight having a correlated color temperature of 7500K (±200K) with illumination of 100 (±20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300K (±200K).

* 3.5.3 Colorfastness. The finished cloth shall show colorfastness to laundering (3 cycles), perspiration, light (40 SFH), and crocking equal to or better than the standard sample when tested as specified in 4.5. As a limit of acceptability, or when no standard sample is available, the finished cloth shall show a minimum of "good" colorfastness to laundering (3 cycles) and perspiration and a minimum of "good" to light (40 SFH), and shall show a dry and wet crocking rating not lower than 3.0.

3.6 Finish. The cloth shall be closely singed, fully mercerized, and have a clear lustrous finish when tested as specified in 4.4.1.

3.6.1 Nonfibrous material. The finished cloth shall not exceed 2.0 percent starch and protein content including chloroform-soluble and water-soluble material when tested as specified in 4.5.

3.7 Dimensional stability. The finished cloth shall not shrink more than 2.0 percent in either direction of the warp or the filling when tested as specified in 4.5. The pre-shrinking process used shall not be identified by name or trademark, either on the cloth, ticket, or package.

3.8 pH. The pH value of the water extract of the finished cloth shall be no less than 5.0 or more than 8.5 when tested as specified in 4.5.

3.9 Seam efficiency. The finished cloth shall have a seam efficiency of not less than 80 percent when tested as specified in 4.5.

3.10 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards (36.4m). Each length shall be put up on full rolls as specified in PPP-P-1134.

3.11 Fiber identification. Each roll of cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Products Identification Act.

3.12 Face identification. The face side of the cloth shall be identified by stamping the word "FACE" on each end of the roll or piece.

3.13 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards ($83.6m^2$), when calculated as specified in Section 4, shall not exceed the established maximum point values.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2, shall be inspected as specified in 4.4.2 for compliance with design, construction, workmanship, appearance and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. In addition, the contractor shall furnish a certificate of compliance for the cotton fiber identification, the carding of the cotton fibers, and the singeing and mercerization of the finished cloth requirements of 3.3.1 and 3.6.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.4.

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4.4.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards (91.4m), the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.4m) only 100 yards (91.4m) shall be examined. All defects as defined in Section 1 of FED-STD-4, which are clearly noticeable at normal inspection distance, 3 feet (0.91m) shall be scored and assigned demerit points as listed in 4.4.2.1.1, except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scale (see 6.4), "E" or 5 (as applicable) for slubs and "B" for knots, shall be scored. No linear yard (increments of 1 yard (0.91m) on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of the total yardage examined exceeds 30.0 points. The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of two or more individual rolls exceeds 45.0 points. If no individual roll exceeds the point level, the lot shall be acceptable with respect to this characteristic. If one roll exceeds the point level, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds the point level. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample X 3600}}{\text{Contracted width of cloth (inches) X total yards inspected}} = \text{Points per 100 square yards } (83.6m^2)$$

4.4.2.1.1 Demerit points. Demerit points shall be assigned as follows.

For defects up to and including 3 inches (7.6cm) in any dimension	- one point
For defects exceeding 3 inches (7.6cm) but not exceeding 6 inches (15.2cm) in any dimension	- two points
For defects exceeding 6 inches (15.2cm) but not exceeding 9 inches (22.9cm) in any dimension	- three points
For defects exceeding 9 inches (22.9cm) in any dimension	- four points

NOTE: The following defects, when present, shall be scored 4 points for each yard (.91m) in which they occur:

Baggy, ridgy or wavy cloth
Width less than specified
Non-uniformity of shade (poor dye penetration, mottled, streaky or cloudy)
Excessive neppiness
Shade defects - shaded side to side, side to center, or end to end
Characteristics of finish, hand, and shade not equal to the standard sample

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4.4.2.2 Examination for length.

4.4.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards (1.82m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.4.2.3 Examination for shade variation. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation. Any roll in the sample exhibiting uneven shade, shade variation side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

4.4.2.4 Examination for face identification, non-identification of preshrinking process, and non-conformance with the Textile Fiber Products Idenfication Act. During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends or placed on the wrong side, contain identification of the preshrinking process by name or trademark on the cloth or ticket, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.4.3 Examination of packaging requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that preservation, packing, and marking comply with Section 5 requirements of this specification.

4.5 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table II shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final results. The sample unit for test purposes shall be four (4) continuous yards (3.64m), full width of the finished cloth. The lot size shall be expressed in units of one yard (0.91m). The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be in accordance with the following:

Lot Size (yards)	Sample Size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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Table II - Test Methods

Characteristic	Requirement Paragraph	Test Method
Yarn ply:		
Warp	3.3.1	Visual 1/
Filling	3.3.1	Visual 1/
Weight	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Weave	3.4.1	Visual 1/
Sulfur content	3.5.1	2020
Colorfastness to:		
Light (40 SFH)	3.5.3	5660 5/
Crocking	3.5.3	AATCC No. 8
Perspiration	3.5.3	5680
Laundering (3 cycles)	3.5.3	5610 2/ 3/
Nonfibrous materials	3.6.1	2611
Dimensional stability	3.7	5550
pH	3.8	2811
Seam efficiency	3.9	5110 4/

1/ One determination per each sample unit and the results reported as "pass" or "fail".

2/ **The specimens must be dried after each of the 3 laundering cycles.**

3/ Only the stain of the cotton fiber shall be rated when evaluating the color transfer cloth.

4/ The needle size shall be 0.032 inch (±0.001)(0.8 mm (±0.025)) across the blade. The thread shall conform to V-T-276, type IA1, top (needle) thread ticket No. 60, 3 ply, and bottom (looper) thread ticket No. 70, 3 ply.

5/ The calibration of the fadeometer and the definition of AATCC Fading Units shall be according to AATCC Test Method 16. Using the formula below 16 to 22 AFU's are acceptable as 20 SFH's.

$$\text{AATCC Fading Units} = \frac{(\Delta E^*)(20)}{1.7}$$

Where (ΔE^*) is the color difference CIELAB* units of color difference.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

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5.1.1 Levels A and C. The cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

* 5.3 Marking. In addition to any special marking required by the contract or order, unit packs, intermediate or exterior containers, and palletized unit loads shall be marked in accordance with MIL-STD-129.

* 5.4 MIL-STD-2073. When specified (see 6.2), Preservation and Packing shall be in accordance with MIL-STD-2073-1 and MIL-STD-2073-2.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cloth covered by this specification is intended for use in a garment worn over the metallized radar frequency protection coverall.

6.2 Acquisition requirements. Acquisition documents should specify the following.

a. Title, number, and date of this specification
b. Color of cloth if other than specified (see 3.5)
c. Width required (see 3.4.2)
d. Minimum length if other than specified (see 3.10)
e. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)
f. When first article sample is required (see 3.2). The item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quality, testing and approval of the first article.
g. Levels of preservation and packing required (see 5.1 and 5.2)
h. When all packaging data is to be found in MIL-STD-2073-1 and MIL-STD-2073-2 (see 5.4).

6.3 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids (see 3.1).

6.4 Fabric defect scales. Sears Fabric Defect Replica Kits are available from:

Sears, Roebuck and Company
Department 817, (ATTN: BSC 23-29)
Sears Tower
Chicago, IL 60684

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6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

* 6.5 Subject term (key word) listing.
Fabric, coverall
Radar frequency protection

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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